

# PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

Anonymous ACL submission

## Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. A useful enterprise Text2Cypher benchmark must therefore be private, refreshable, executable, and sensitive to graph-specific semantics such as relationship direction, literal grounding, and schema vocabulary. We present PIPE-Cypher, a local-model pipeline for generating natural-language-to-Cypher benchmarks inside an organization’s compute boundary. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained generation, deterministic Cypher governance, execution validation, diversity controls, privacy redaction, and LLM-judge review. In live public-proxy studies with local Qwen3.5-9B generation and judging, PIPE-Cypher exports 3,000 accepted FinBench/SNB examples, completes three audited ablation suites, calibrates an automated judge, onboards ICIJ Offshore Leaks, and evaluates 11 completed local downstream models. The resulting benchmark is difficult zero-shot; the strict no-signature few-shot control shows schema-specific examples can help compatible model families, while same-signature example banks are reported only as an operational upper bound.

## 1 Introduction

Property graphs are attractive in enterprise settings because the facts of interest are often relational paths: account transfers, identity entitlements, access chains, customer interactions, supplier dependencies, and fraud rings. Cypher gives analysts a compact language for these patterns. As LLMs become natural-language interfaces for graph analytics, an organization cannot evaluate a Text2Cypher system only on public schemas; it needs to know whether the model handles its labels, relationship directions, values, governance rules, and recurring operational questions.

This changes the benchmark problem. A static public dataset is valuable for shared comparison, but it cannot contain a bank’s account taxonomy, an identity team’s permission graph, or the exact categorical values that make a query answerable. It also cannot refresh when schemas change. Industry teams therefore need a benchmark factory: a repeatable way to turn a live property graph into a balanced, executable, privacy-aware NL-to-Cypher benchmark without sending sensitive schema or values to paid generation APIs.

PIPE-Cypher addresses this setting. The pipeline profiles a target graph, grounds question slots through reverse Cypher execution, constrains local-model generation, validates and repairs generated Cypher through deterministic gates, and uses a local LLM judge only after execution evidence is available. Its design treats Cypher correctness as a governed artifact rather than a prompt preference: relationship direction, read-only safety, exact literal use, categorical values, contextual return columns, and `RETURN DISTINCT` are checked or normalized before examples are accepted.

**Contributions.** We make four contributions: (1) an end-to-end local-model benchmark-factory workflow for organization-run NL-to-Cypher evaluation; (2) outcome-aware reverse grounding plus deterministic Cypher governance for read-only safety, directionality, exact literals, categorical values, contextual returns, and conservative rewrite boundaries; (3) a scaled public-proxy evaluation over FinBench, SNB, and ICIJ with ablations, judge calibration, privacy/redaction audits, and an 11-model completed local transfer stress test; and (4) reproducibility artifacts for onboarding, value sampling, benchmark refresh, evidence packaging, and appendix-level auditability.

## 2 Related Work

Recent Text2Cypher resources, including Mind the Query (Chauhan et al., 2025), SyntheT2C (Zhong et al., 2025), Auto-Cypher (Tiwari et al., 2025), Text2Cypher (Ozsoy et al., 2025), Cypher-Bench (Feng et al., 2025), and the public Text2Cypher-2024 corpus (Neo4j, 2024), show that high-quality Cypher data requires schema grounding, execution checks, verification, and complexity-aware evaluation. PIPE-Cypher does not treat template filling or execution filtering as new in isolation. Its technical delta is the enterprise benchmark-factory composition: outcome-aware reverse grounding, deterministic Cypher governance, read-only deployment, local judge calibration, privacy/value policies, refreshable exports, and evidence ledgers.

| Mechanism              | Prior Text2Cypher use                                     | PIPE-Cypher delta                                                                                  |
|------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Template/LLM synthesis | SyntheT2C and Auto-Cypher generate Question-Cypher pairs. | Outcome-aware reverse grounding binds slots through live Cypher before NL realization.             |
| Execution validation   | Auto-Cypher and Mind the Query retain executable queries. | Execution is one gate in a ledger with direction, read-only, value, non-empty, and judge evidence. |
| Schema/value checks    | Mind the Query validates schema, runtime, and values.     | Checks are packaged for private refresh with configurable value sampling and redacted exports.     |
| Logical review         | Mind the Query uses human logical validation.             | Human labels calibrate a local LLM judge; they are not a generation gate.                          |
| Benchmark artifact     | Public static datasets and tuning corpora.                | Organization-run benchmark factory with manifests, refresh, and provenance audits.                 |

Table 1: Prior mechanism comparison. PIPE-Cypher does not claim that execution filtering or template filling are new by themselves; the contribution is the enterprise benchmark-factory composition with outcome-aware grounding, deterministic Cypher governance, local judging, privacy policy, and audit ledgers.

Mind the Query is especially relevant because it reports an Industry Track Text2Cypher dataset with schema, runtime, value, and human logical valida-

tion. PIPE-Cypher borrows the reviewer-facing discipline of those checks, but changes the target artifact: instead of releasing one static dataset or tuning corpus, it studies the process of generating, refreshing, auditing, and redacting benchmarks for a target graph, local model endpoint, privacy policy, and benchmark refresh cycle.

Text-to-SQL benchmarks such as Spider 2.0 (Lei et al., 2024) and BIRD (Li et al., 2023) have pushed text-to-query evaluation toward realistic database tasks and execution-based scoring, while execution-guided decoding shows why query execution is useful semantic feedback rather than a cosmetic check (Wang et al., 2018). Recent residual-skill Text-to-SQL work further shows that optimizing complementary agent skills on residual failures can improve selected accuracy across SQL dialects (Zhu et al., 2026). Graph query generation adds different failure modes: relationship direction can invert the meaning of a query, node and relationship properties live in different namespaces, and path-shaped questions can be syntactically valid while semantically ungrounded. CIKM AutoQuery (Zheng et al., 2024) motivates treating workload generation as a separate object of study from downstream model quality. LDBC FinBench (Qi et al., 2023) and SNB (Puroja et al., 2023) provide public graph workloads with financial and social-network structure, while ICIJ Offshore Leaks gives an additional public finance/compliance onboarding check.

For benchmark quality, lexical diversity alone is not enough. PIPE-Cypher combines text-generation diagnostics such as Distinct-n (Li et al., 2016) and self-BLEU-style redundancy checks (Zhu et al., 2018) with text-to-query structure metrics: schema coverage, relationship/property coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells. For subset construction, PIPE-Cypher uses an MMR-style novelty objective (Carbonell and Goldstein, 1998) over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens. These metrics make a practical distinction that matters to enterprise users: a benchmark can be syntactically diverse while still overusing the same values or query templates.

### 3 Method

PIPE-Cypher has six stages: schema profiling, workload planning, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. The central design choice is to make answerability and governance executable. Prompts can ask a model to respect relationship directions or exact literals, but accepted examples must prove those properties through schema checks, parser-style structure extraction, live execution, and judge review.

Schema profiling records labels, relationship types, properties, observed directions, and bounded low-cardinality categorical values. Workload planning targets eight operational categories: simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k. Reverse grounding runs read-only Cypher to bind slots to values that actually produce rows, reducing the common synthetic-data failure where a plausible question has no answer in the graph.

The deterministic layer enforces read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. Live execution uses read-only credentials and read-access sessions; token-level write rejection is therefore a preflight gate rather than the only safety mechanism. A lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

### 4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Reported benchmark generation and judge review use a local Qwen3.5-9B endpoint behind a vLLM/OpenAI-compatible interface. Downstream evaluation separately tests 11 completed locally served model

checkpoints spanning general instruction, code-tuned, Cypher-tuned, and Text2Cypher-tuned families. This keeps the study aligned with an enterprise deployment pattern in which benchmark generation and evaluation run inside the organization’s compute boundary without paid generation APIs.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For generation, PIPE-Cypher uses seeded workload templates with deterministic reverse-binding queries and a mixed mode that prepends proven workload seeds to LLM-proposed templates. Every run records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

Inspired by Mind the Query’s prompt-setting analysis, the implementation exposes prompt profiles for schema-only, instructions-only, examples-only, examples-plus-instructions, and full governed PIPE-Cypher generation. These profiles are configured as ablation variants and must pass the same target-size and paper-readiness standards before any result is reported.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B prompts within context limits on larger schemas.

The deterministic Cypher layer uses production-derived constraints: schema-only prompting, exact matching, relationship direction discipline, RETURN DISTINCT, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite bound-

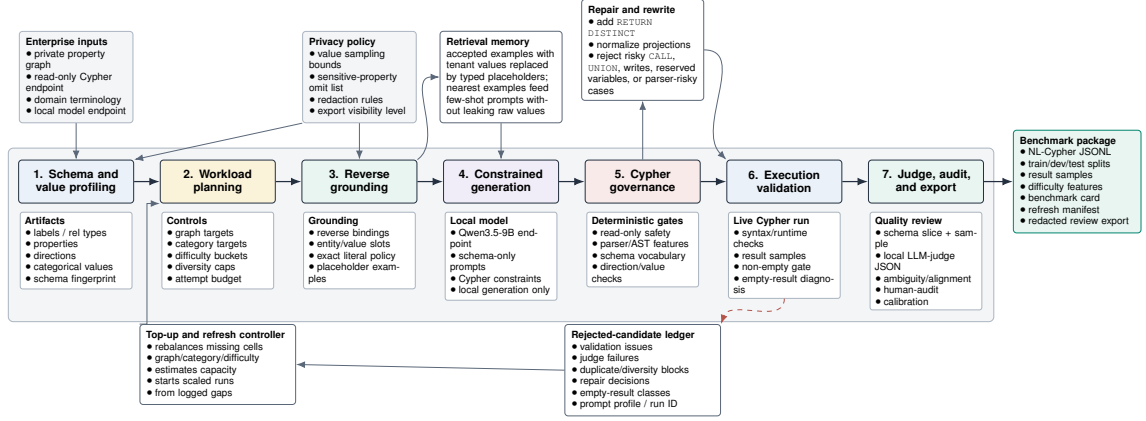


Figure 1: PIPE-Cypher benchmark generation pipeline. The key industry additions beyond static Text2Cypher dataset construction are privacy/value policies, reverse grounding, Cypher governance, execution diagnostics, local judge calibration, and benchmark export/refresh.

| Gate                              | Evidence checked                                           | Failure mode caught                       |
|-----------------------------------|------------------------------------------------------------|-------------------------------------------|
| Read-only safety                  | blocked Cypher write/admin tokens                          | destructive or operational queries        |
| Schema validity                   | labels, relationship types, properties, categorical values | hallucinated graph vocabulary or values   |
| Direction validity                | observed relationship directions                           | reversed or invalid traversals            |
| Cypher analysis and normalization | structure extraction, rewrite safety, RETURN DISTINCT      | unsafe rewrites or duplicate/noisy rows   |
| Question constraints              | quoted values use exact literals                           | fuzzy matching of exact user values       |
| Execution                         | query runs and returns rows                                | syntactic validity without answerability  |
| LLM judge                         | ambiguity, semantic alignment, schema use                  | executable but semantically weak examples |

Table 2: PIPE-Cypher candidate acceptance gates.

aries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as UNION, CALL, UNWIND, WHERE EXISTS, multiple WHERE clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

For enterprise onboarding, PIPE-Cypher includes a deployment template for a company’s own graph, read-only credentials, local model endpoint, schema introspection, privacy policy, dry run, scaled run, audit, and redacted export. This paper validates the pattern on three public proxy graphs rather than a proprietary tenant deployment. Configurable value-sampling policies bound which low-cardinality graph values enter schema prompts, and redacted exports replace quoted literals, entity values, and string-valued result samples with stable placeholders for broader internal review. The ICIJ

| Graph    | Target/category | Binding limit | Min seed capacity | All categories meet target |
|----------|-----------------|---------------|-------------------|----------------------------|
| FinBench | 250             | 300           | 300               | Yes                        |
| SNB      | 125             | 200           | 200               | Yes                        |

Table 3: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

onboarding run further motivated schema-derived sparse-category templates: PIPE-Cypher now derives relationship-count, anti-join, and top-k templates from observed labels, relationship directions, and safe low-cardinality properties, then grounds slot values with outcome-aware reverse Cypher.

## 5 Experiments

**Research questions.** We evaluate PIPE-Cypher around four questions that matter for an industry benchmark generator: RQ1, can a local-model pipeline produce a balanced executable benchmark over live property graphs? RQ2, do Cypher-specific governance and grounding steps make generation reliable at scale? RQ3, does the resulting benchmark expose meaningful downstream

| Graph    | Candidates | Accepted | Acceptance | Categories at target |
|----------|------------|----------|------------|----------------------|
| FinBench | 3,405      | 2,000    | 0.587      | 8/8                  |
| SNB      | 1,520      | 1,000    | 0.658      | 8/8                  |
| Total    | 4,925      | 3,000    | 0.609      | 16/16                |

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

Text2Cypher failures rather than merely checking syntax? RQ4, can the same machinery onboard a new public enterprise-style graph without hard-coding FinBench or SNB?

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB, balanced over eight workload categories. We report three completed FinBench/SNB ablation suites: target-50, corrected target-100, and a seed-17 target-50 repeat. Downstream Text2Cypher evaluation uses live execution accuracy and answer-set F1 as primary metrics; reference-based text metrics are supported for debugging only and reported in the appendix. We additionally report ICIJ Offshore Leaks as a third public finance/compliance onboarding proxy.

## 6 Results

**RQ1: executable benchmark generation.** The full live run produced 3,000 accepted examples from 4,925 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

The accepted records are exported with stable identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash; the appendix gives the full artifact distribution.

**Diversity and residual concentration.** We report diversity as an audit signal, not as a slogan. The full export has perfect category balance, near-perfect difficulty balance, 1,115 unique grounded entity values, and exact quotation of grounded values in 82.6% of examples with entity bindings. Query-signature diversity remains low because seeded graph-grounded templates do substantial work, but a diversity-governed selector improves structural coverage, adjusted Distinct-2, query-signature ratio, and property coverage at the same graph/category target (Appendix Table 22). Thus PIPE-Cypher produces a balanced executable benchmark while making residual template and

| Split          | Examples | Parse | Schema | Exec. success | Exec. acc. | Answer F1 |
|----------------|----------|-------|--------|---------------|------------|-----------|
| Live full test | 296      | 0.963 | 0.916  | 0.611         | 0.189      | 0.189     |

Table 5: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

value concentration visible for refresh.

**RQ2: governed generation.** The full-run failure taxonomy is concentrated in duplicate/diversity controls and empty execution results; only 2/4,925 candidates were schema-invalid after governance. A rewrite audit found that all paper-run candidates were already identical to their normalized Cypher, so accepted examples do not depend on semantics-changing rewrites. The ablations are therefore a reliability study of the execution-grounded core: in the target-100 suite, every non-unconstrained graph/setting cell reached all eight category targets, while unconstrained generation did not provide balanced executable coverage. Across the three evidence-ready suites, target-normalized coverage is 1.000 for every non-unconstrained cell.

**Judge calibration.** An 80-row post-hoc human audit sampled accepted and rejected candidates across both graphs and all categories. Agreement is 80.0%, Cohen’s  $\kappa = 0.60$ , judge precision/specificity are 1.00, recall is 0.714, and no false accepts appear in the sample. The judge is conservative and protects accepted-example quality; human labels calibrate the gate but do not participate in generation.

**RQ3: downstream stress test.** We evaluate downstream Text2Cypher by giving each model schema text and the question, then scoring live execution on FinBench or SNB. The purpose is not to present a strong baseline; it is to test whether PIPE-Cypher exposes plausible but wrong Cypher. In an 11-model completed local transfer suite, zero-shot execution accuracy ranges from 0.000 to 0.203 (mean 0.036). The primary few-shot generalization result is the scored no-signature control, which excludes exact query-signature matches and near-duplicate questions and raises mean accuracy to 0.200. Ordered and random same-category example banks reach 0.269/0.267, but are reported as operational upper bounds because they often share query signatures (Appendix Tables 16–18).

**RQ4: third-graph onboarding.** ICIJ onboarding reaches 800 accepted examples from 983 candidates on a 2.0M-node, 3.3M-edge public finance/compliance graph, with 100 examples in every category. This is third-public-proxy evidence,



not proof of private-tenant coverage, but it exercises schema-derived relationship-count, anti-join, and top-k templates beyond the two LDBC study workloads.

## 7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The artifact records schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Teams can refresh after schema changes, compare models on the same held-out split, inspect failures by category, build a schema-specific question-answer demonstration bank for retrieval or adaptation, and export redacted review packets for governance. Deployment requires read-only graph credentials, schema introspection, a bounded value policy, a local endpoint, a dry run, scaled generation, judge calibration, and redacted export. The FinBench/SNB/ICIJ study validates this pattern on public proxy graphs while making the proprietary-tenant gap explicit.

## 8 Conclusion

PIPE-Cypher reframes Text2Cypher benchmarking as a private, repeatable enterprise workflow. The main lesson is that benchmark generation improves when Cypher constraints become executable gates: reverse grounding makes questions answerable, deterministic governance catches unsafe or schema-invalid queries, execution exposes empty or brittle candidates, diversity diagnostics reveal concentration, and a calibrated local judge provides a conservative semantic filter. This produces a benchmark that is not only larger, but more useful: it is balanced, auditable, refreshable, and able to reveal downstream model failures that syntax-only evaluation would hide.

## Limitations

Execution validity does not guarantee semantic correctness. The completed 80-row, single-audit-sheet calibration suggests a conservative judge with no observed false accepts in the labeled sample, but the confidence interval is necessarily wider than the point estimate and larger multi-annotator audits may reveal additional failure modes. FinBench, SNB, and ICIJ are public enterprise-style proxies rather than a proprietary tenant graph, so the study

tests the onboarding pattern but not every deployment constraint of a real organization. The full export is balanced by graph, category, and difficulty, yet query-signature diagnostics show residual template concentration from the seeded, execution-grounded generation strategy. PIPE-Cypher deliberately disallows or skips risky Cypher constructs such as writes, undirected relationships, UNION, CALL, UNWIND, and parser-risky rewrites; this is a safe benchmark-generation subset, not complete coverage of every production Cypher idiom. The downstream few-shot result should be read as graph-specific example-bank conditioning: the no-signature control is the leakage-aware result, while ordered and random same-category demonstrations are upper-bound operating conditions that often share query signatures. Tenant-specific fine-tuning remains an engineering path enabled by the artifact, not a completed deployment claim in this paper. The redaction audit checks exact residuals for known value-bearing strings, but it is not a full PII classifier and does not make schema names confidential.

## Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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| Setting                | Value                               |
|------------------------|-------------------------------------|
| Accepted examples      | 3,000                               |
| Primary graph          | LDBC FinBench, 2,000 examples       |
| Secondary graph        | LDBC SNB, 1,000 examples            |
| Categories             | 8 balanced categories, 375 each     |
| Generation/judge model | Local Qwen3.5-9B                    |
| Execution backend      | Neo4j Community, two live databases |
| Judge audit packet     | 80 sampled accepted/rejected pairs  |

Table 6: Full live experimental setup used for the generated benchmark artifact.

| Export artifact     | Examples | FinBench | SNB   | Train | Dev | Test |
|---------------------|----------|----------|-------|-------|-----|------|
| Live full benchmark | 3,000    | 2,000    | 1,000 | 2,408 | 296 | 296  |

Table 7: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash `cf274344be2abe7e`.

## A Governed Generation Evidence

The appendix is organized as a claim-evidence trace. Each table or figure appears next to the paragraph that interprets it so the reader does not need to reconstruct the argument from a block of floats. The first claim is reliability: once reverse grounding and deterministic Cypher governance are in place, PIPE-Cypher fills every planned graph/category cell at target scale.

### A.1 Benchmark Artifact Summary

Tables 6, 7, and 8 give the setup, export manifest summary, and gate/distribution details that are compressed in the six-page main body.

### A.2 Target-100 Stress Baseline

Figure 2 and Table 9 are the detailed evidence behind the compact RQ2 discussion. The unconstrained rows are intentionally shown as a stress baseline: plausible raw local-model generations do not produce balanced, executable benchmark coverage. The governed variants, in contrast, fill every target cell on both graphs.

| Setting           | Graph    | Attempts | Records | Accepted | Acceptance | Categories at target |
|-------------------|----------|----------|---------|----------|------------|----------------------|
| Unconstrained LLM | FinBench | 422      | 422     | 200      | 0.474      | 2/8                  |
| Unconstrained LLM | SNB      | 2,000    | 2,000   | 50       | 0.025      | 0/8                  |
| Reverse-only      | FinBench | 815      | 815     | 800      | 0.982      | 8/8                  |
| Reverse-only      | SNB      | 820      | 820     | 800      | 0.976      | 8/8                  |
| Validators+repair | FinBench | 833      | 833     | 800      | 0.960      | 8/8                  |
| Validators+repair | SNB      | 824      | 824     | 800      | 0.971      | 8/8                  |
| No retrieval      | FinBench | 819      | 819     | 800      | 0.977      | 8/8                  |
| No retrieval      | SNB      | 809      | 809     | 800      | 0.989      | 8/8                  |
| No rewrite        | FinBench | 826      | 826     | 800      | 0.969      | 8/8                  |
| No rewrite        | SNB      | 834      | 834     | 800      | 0.959      | 8/8                  |
| No LLM judge      | FinBench | 812      | 812     | 800      | 0.985      | 8/8                  |
| No LLM judge      | SNB      | 828      | 828     | 800      | 0.966      | 8/8                  |
| Full PIPE-Cypher  | FinBench | 824      | 824     | 800      | 0.971      | 8/8                  |
| Full PIPE-Cypher  | SNB      | 824      | 824     | 800      | 0.971      | 8/8                  |

Table 9: Live target-100 ablation evidence with local Qwen3.5-9B. Governed graph runs target 100 accepted examples per category; the unconstrained row is a stress baseline reported with explicit attempt accounting.

| Artifact property                   | Value                     |
|-------------------------------------|---------------------------|
| Categories                          | 8 balanced categories     |
| FinBench/category                   | 250                       |
| SNB/category                        | 125                       |
| Difficulty split                    | 1,569 easy / 1,431 medium |
| Used labels / rel. types            | 16 / 17                   |
| Read/syntax/schema/exec/judge gates | 3,000/3,000               |

Table 8: Distribution and gate summary for the exported full benchmark artifact.

### A.3 Gate-Level Quality

Yield alone is not the main result; a benchmark factory must explain which gates are doing useful work. Table 10 and Figure 3 therefore report read-only, syntax, schema, execution, and judge/post-hoc rates for the same target-100 suite. The quality story is that read-only and syntax checks are saturated after governance, while execution and semantic judging expose the remaining differences.

| Setting           | Graph    | Read-only | Syntax | Schema | Exec. | Judge/post-hoc |
|-------------------|----------|-----------|--------|--------|-------|----------------|
| Unconstrained LLM | FinBench | 1.000     | 1.000  | 0.806  | 0.723 | 0.557          |
| Unconstrained LLM | SNB      | 1.000     | 0.998  | 0.599  | 0.478 | 0.144          |
| Reverse-only      | FinBench | 1.000     | 1.000  | 0.982  | 0.982 | 0.982          |
| Reverse-only      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.976          |
| Validators+repair | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.960          |
| Validators+repair | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.971          |
| No retrieval      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.977          |
| No retrieval      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.989          |
| No rewrite        | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.969          |
| No rewrite        | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.959          |
| No LLM judge      | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.985          |
| No LLM judge      | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.966          |
| Full PIPE-Cypher  | FinBench | 1.000     | 1.000  | 1.000  | 1.000 | 0.971          |
| Full PIPE-Cypher  | SNB      | 1.000     | 1.000  | 0.998  | 0.998 | 0.971          |

Table 10: Quality-gate rates for the live target-100 ablation suite. Rates are computed over all generated records in each graph/setting; for no-judge settings, the judge column is a post-hoc scoring diagnostic.

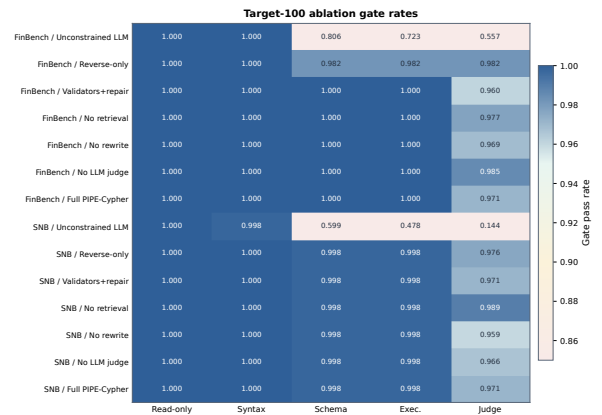


Figure 3: Gate-rate heatmap for the audited target-100 ablation suite. Deterministic read-only and syntax gates are saturated; execution and judge rates expose the remaining quality differences across graph and pipeline variants.

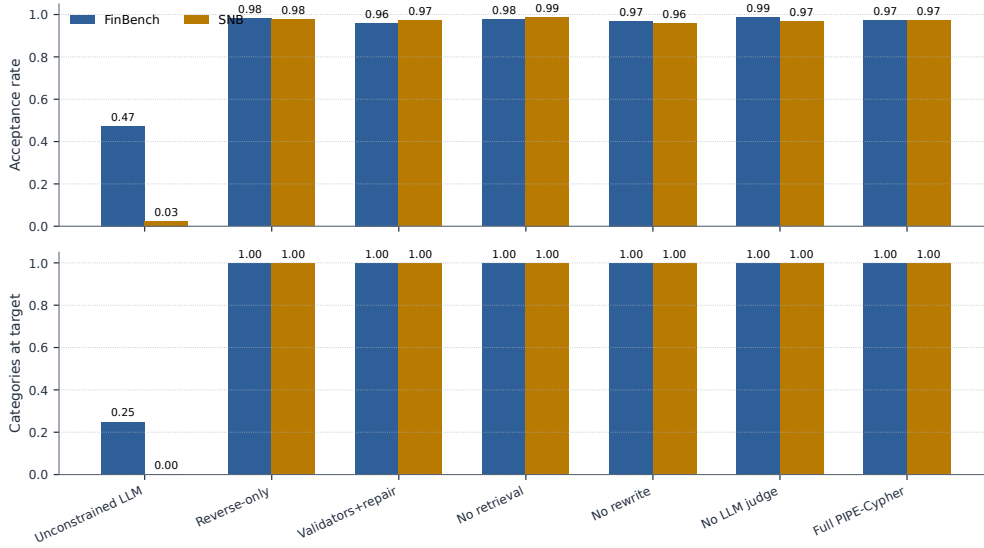


Figure 2: Target-100 ablation yield on FinBench and SNB. Unconstrained local generation is reported as a stress baseline with explicit attempt accounting; the execution-grounded governed variants reach all eight workload-category targets on both graphs. The takeaway is reliability of the grounded governance core, not that every optional module is needed for yield.

#### A.4 Repeated-Suite Stability

The same conclusion holds across target sizes and seeds. The repeated-suite comparison normalizes by each suite’s planned target, so the table and figure show stability rather than rewarding larger raw counts. Full PIPE-Cypher and the governed ablations reach complete target coverage on both graphs, supporting the claim that the system behaves like a repeatable benchmark factory rather than a one-off generation script.

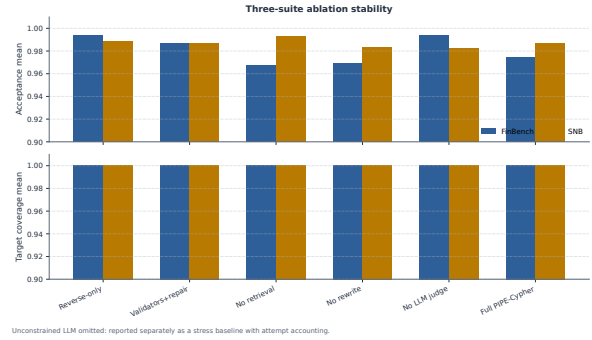


Figure 4: Three-suite ablation stability over the original target-50 suite, corrected target-100 suite, and seed-17 target-50 repeat. Target-normalized coverage stays at 1.000 for all non-unconstrained cells.

| Setting           | Graph    | Suites | Target cov. | Acceptance  | Cat. target | Exec. | Judge |
|-------------------|----------|--------|-------------|-------------|-------------|-------|-------|
| No LLM judge      | FinBench | 3/3    | 1.000       | 0.994±0.008 | 1.000       | 1.000 | 0.994 |
| No retrieval      | FinBench | 3/3    | 1.000       | 0.967±0.031 | 1.000       | 1.000 | 0.967 |
| No rewrite        | FinBench | 3/3    | 1.000       | 0.969±0.023 | 1.000       | 1.000 | 0.969 |
| Full PIPE-Cypher  | FinBench | 3/3    | 1.000       | 0.975±0.016 | 1.000       | 1.000 | 0.975 |
| Reverse-only      | FinBench | 3/3    | 1.000       | 0.994±0.011 | 1.000       | 0.994 | 0.994 |
| Validators+repair | FinBench | 3/3    | 1.000       | 0.987±0.023 | 1.000       | 1.000 | 0.987 |
| No LLM judge      | SNB      | 3/3    | 1.000       | 0.982±0.015 | 1.000       | 0.996 | 0.982 |
| No retrieval      | SNB      | 3/3    | 1.000       | 0.993±0.004 | 1.000       | 0.996 | 0.993 |
| No rewrite        | SNB      | 3/3    | 1.000       | 0.983±0.021 | 1.000       | 0.996 | 0.983 |
| Full PIPE-Cypher  | SNB      | 3/3    | 1.000       | 0.987±0.014 | 1.000       | 0.996 | 0.987 |
| Reverse-only      | SNB      | 3/3    | 1.000       | 0.989±0.011 | 1.000       | 0.996 | 0.989 |
| Validators+repair | SNB      | 3/3    | 1.000       | 0.987±0.014 | 1.000       | 0.996 | 0.987 |

Table 11: Target-size and repeated-seed ablation sensitivity. Target coverage normalizes accepted examples by each suite’s planned graph/category target, so target-50 and target-100 suites can be compared without treating larger raw counts as quality gains. Unconstrained local generation is excluded from this stability table and reported separately as the attempt-logged stress baseline in Table 9.

#### A.5 Rejected Candidate Taxonomy

The rejection taxonomy explains where the remaining work occurs. After deterministic Cypher governance, schema-invalid candidates are rare; most rejected candidates are duplicate/diversity blocks from recovery or empty-result candidates caught before export. This is the behavior an enterprise deployment wants: invalid or brittle examples remain in the ledger, not in the benchmark.

| Failure bucket              | Count | Share of rejected |
|-----------------------------|-------|-------------------|
| Diversity/duplicate control | 1,366 | 0.710             |
| Empty result                | 486   | 0.252             |
| Judge semantic reject       | 67    | 0.035             |
| Execution error             | 4     | 0.002             |
| Schema invalid              | 2     | 0.001             |

Table 12: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

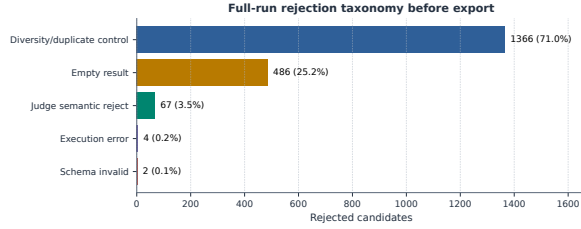


Figure 5: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after deterministic governance.

## B Benchmark Artifact and Third-Graph Onboarding

### B.1 Export Balance

The exported artifact is designed to make scale and balance visible. Figure 6 shows the planned 2:1 FinBench/SNB mix, exact category balance, and near-even difficulty split. This is the artifact-level evidence for the paper’s core claim: PIPE-Cypher produces a benchmark that is balanced by design rather than sampled opportunistically.

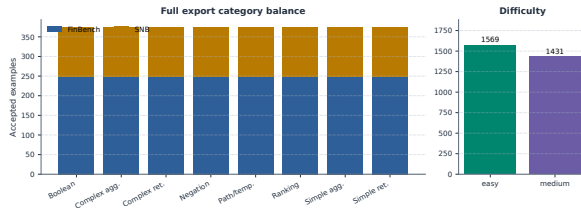


Figure 6: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

### B.2 Graph Scale and Schema Variety

Table 13 anchors the study graphs before introducing the third-graph onboarding audit. FinBench and SNB are the controlled LDBC study workloads; ICIJ is the public finance/compliance graph

used to test whether the same onboarding machinery works outside the two original schemas.

| Graph               | Nodes     | Relationships | Labels | Rel. types | Paper status     |
|---------------------|-----------|---------------|--------|------------|------------------|
| FinBench            | 10,006    | 57,622        | 5      | 9          | reported         |
| SNB                 | 34,735    | 70,842        | 14     | 15         | reported         |
| ICIJ Offshore Leaks | 2,016,523 | 3,339,267     | 5      | 14         | onboarding audit |

Table 13: Study graph statistics. ICIJ Offshore Leaks is used as a public third-graph onboarding audit for arbitrary finance/compliance schemas beyond the two LDBC study workloads.

### B.3 ICIJ Onboarding

Table 14 and Figure 7 report the third-graph result. ICIJ matters because it forced schema-derived sparse-category augmentation rather than relying on preauthored FinBench/SNB templates. The run reaches all eight category targets while preserving the same validation and audit standards.

| ICIJ onboarding property      | Value                                    |
|-------------------------------|------------------------------------------|
| Graph nodes / relationships   | 2,016,523 / 3,339,267                    |
| Labels / relationship types   | 5 / 14                                   |
| Generated / accepted          | 983 / 800                                |
| Acceptance rate               | 0.814                                    |
| Categories at target          | 8/8                                      |
| Paper audit                   | ready                                    |
| Sparse schema-derived accepts | complex agg. 97, negation 28, ranking 98 |

Table 14: ICIJ Offshore Leaks third-graph onboarding audit. The public finance/compliance graph tests arbitrary-schema generation beyond the two LDBC study workloads; raw values remain outside the paper artifacts.

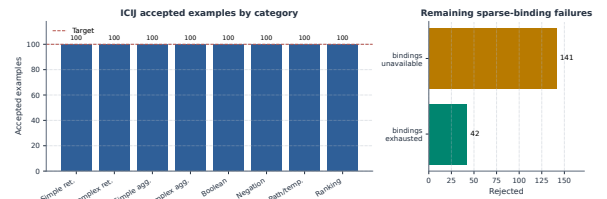


Figure 7: ICIJ Offshore Leaks onboarding audit. Schema-derived sparse-category templates recover/balanced target-100 coverage on a public finance/compliance graph beyond the two LDBC workloads.

### B.4 Category Crosswalk

Table 15 connects PIPE-Cypher’s eight categories to the simpler retrieval/aggregation/evaluation-query taxonomy used in Mind the Query. The purpose is to make comparison easy while showing the extra enterprise workload classes: negation, temporal/path reasoning, and ranking/top-k.

| PIPE-Cypher category      | Closest MTQ category | Added enterprise distinction                                 |
|---------------------------|----------------------|--------------------------------------------------------------|
| Simple retrieval          | SR                   | Direct node/edge lookup with exact filters.                  |
| Complex retrieval         | CR                   | Multi-hop or multi-pattern retrieval.                        |
| Simple aggregation        | SA                   | Single aggregation such as count/min/max/average.            |
| Complex aggregation       | CA                   | Grouped or multi-stage aggregation over graph neighborhoods. |
| Boolean existence         | EQ                   | Precise yes/no or existence answer.                          |
| Negation/difference       | CR                   | Absence, anti-join, or difference query.                     |
| Path/temporal transaction | CR/CA                | Temporal or path-oriented transaction neighborhood.          |
| Ranking/top-k             | SA/CA                | Ordered top-k query with explicit limit.                     |

Table 15: Category crosswalk to Mind the Query. PIPE-Cypher keeps the familiar retrieval/aggregation/evaluation-query structure while adding enterprise workloads such as negation, temporal paths, and ranking.

## C Downstream Transfer and Example-Bank Utility

### C.1 Transfer Controls

The downstream experiment is a benchmark-utility stress test. A useful enterprise benchmark should not merely reward syntactically valid Cypher; it should reveal when a model produces a query that parses, mentions plausible schema elements, and even executes, but answers the wrong operational question. The Qwen3.5-9B result has that shape: parse validity is 0.963 and schema validity is 0.916, while exact execution accuracy is only 0.189 on the full held-out split. This gap is evidence that PIPE-Cypher is measuring semantic graph-query competence rather than only query formatting.

The multi-model transfer suite compares local general instruction, code-tuned, Cypher instruction, and Text2Cypher-finetuned models while keeping the schema prompt, evaluation backend, split, and execution metrics fixed. Figure 8 is the key result: zero-shot transfer is weak, but accepted graph-specific examples become a high-value private question-answer bank. Demonstration-bank controls close much of the gap for compatible model families such as Qwen, Qwen-Coder, and one Gemma Text2Cypher LoRA, while still revealing brittle public fine-tuned checkpoints.

Table 17 reports checkpoint-level uncertainty for the same controls. The interval is deliberately conservative because the unit is model checkpoint rather than question row; it supports the narrower claim that example-bank gains are model-family dependent, not universal.

| Control                   | Mean acc. | $\Delta$ vs zero | 95% $\Delta$ CI | Models improved |
|---------------------------|-----------|------------------|-----------------|-----------------|
| Ordered same-category     | 0.269     | 0.233            | [0.000, 0.481]  | 3/11            |
| Scored no-signature       | 0.200     | 0.163            | [0.000, 0.335]  | 3/11            |
| Random same-category mean | 0.267     | 0.231            | [0.000, 0.464]  | 3/11            |

Table 17: Model-level paired bootstrap uncertainty for downstream few-shot controls. The unit of resampling is the local checkpoint, not an individual question, so the interval is a conservative check on whether gains are broad across model families. Zero-shot mean execution accuracy is 0.036.

Table 18 is placed with the transfer result because it defines the correct interpretation: ordered and random same-category demonstrations are useful example-bank conditions, while the scored no-signature condition is the stricter leakage-aware control.

| Selection mode | Rows | Demos | Sig. match | High sim. | Mean sim. |
|----------------|------|-------|------------|-----------|-----------|
| Ordered        | 296  | 1480  | 0.866      | 0.389     | 0.825     |
| No-signature   | 296  | 1480  | 0.000      | 0.000     | 0.585     |
| Random         | 296  | 1480  | 0.854      | 0.409     | 0.824     |

Table 18: Few-shot leakage controls for downstream demonstration-bank evaluation. The held-out split has 0 exact train/test question overlaps and 289 train/test query-signature overlaps; selection-mode rates show how often retrieved demonstrations share the test query signature or exceed the 0.90 normalized-question similarity threshold.

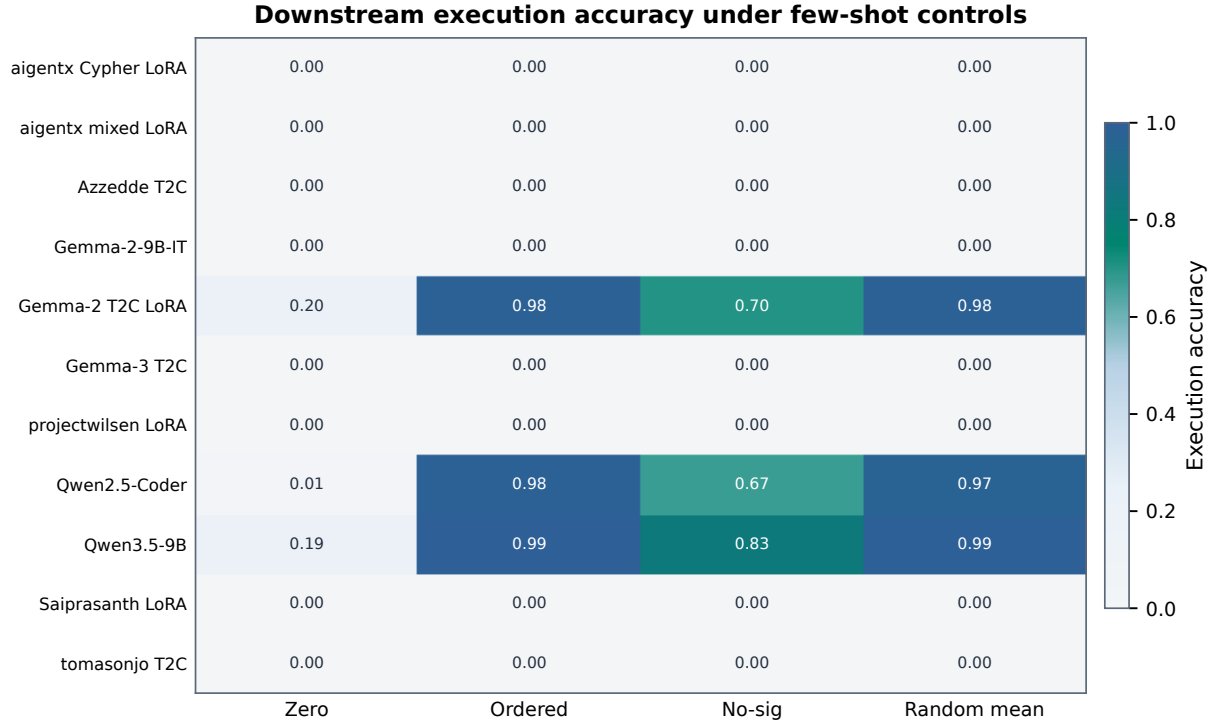


Figure 8: Execution accuracy for 11 completed local downstream models under zero-shot and few-shot control modes. The heatmap highlights both findings: schema-specific examples can sharply help compatible models, and several public fine-tuned checkpoints still fail under enterprise-style Cypher prompting.

| Model                                       | Tuning                 | Zero  | Ordered | No-sig | Random $\mu$ | Random $\sigma$ | Best            |
|---------------------------------------------|------------------------|-------|---------|--------|--------------|-----------------|-----------------|
| aigentx/Llama-3.1-8B Cypher LoRA            | Cypher LoRA            | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| aigentx/Llama-3.1-8B Cypher mixed LoRA      | Cypher mixed LoRA      | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Azzedde/Llama3.1-8b-text2cypher             | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Gemma-2-9B-IT                               | general instruction    | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| neo4j/Gemma-2-9B Text2Cypher LoRA           | Text2Cypher LoRA       | 0.203 | 0.983   | 0.699  | 0.981        | 0.009           | ordered (0.983) |
| neo4j/Gemma-3-4B Text2Cypher                | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| projectwilsen/Llama-3.1-8B Text2Cypher LoRA | Text2Cypher LoRA       | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| Qwen2.5-Coder-7B-Instruct                   | code instruction       | 0.007 | 0.983   | 0.669  | 0.972        | 0.002           | ordered (0.983) |
| Qwen3.5-9B                                  | general instruction    | 0.189 | 0.993   | 0.828  | 0.986        | 0.007           | ordered (0.993) |
| Saiprasanth15/Llama-3.1-8B Text2Cypher LoRA | Text2Cypher LoRA       | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |
| tomasonjo/text2cypher-demo-16bit            | Text2Cypher fine-tuned | 0.000 | 0.000   | 0.000  | 0.000        | 0.000           | no gain         |

Table 16: Few-shot demonstration-bank controls for local downstream Text2Cypher evaluation over completed 296-example local-model runs. Ordered uses the deterministic same-graph, same-category example bank; scored excludes exact query-signature matches and near-duplicate questions; random reports the mean and standard deviation across seeds 13, 17, and 23. “No gain” means no few-shot control exceeded that model’s zero-shot execution accuracy.



C.2 Downstream Failure Modes and Supplementary Metrics

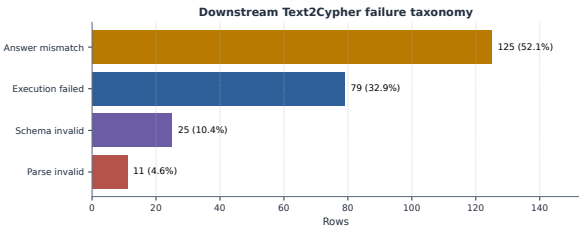


Figure 9: Downstream Text2Cypher failure taxonomy for the local Qwen3.5-9B baseline on the full test split. Exact matches are excluded so the chart exposes whether misses come from invalid Cypher, failed execution, or semantically wrong executable queries.

Table 20 is evaluator infrastructure rather than a headline result. It documents reference-based text metrics that are useful when debugging answer rendering or near-match behavior. The paper’s correctness claims remain grounded in execution accuracy and answer-set F1.

The failure taxonomy is included next to the transfer results because it explains why the benchmark is useful operationally. Incorrect rows are not all the same: some outputs fail to parse, some mention invalid schema, some fail at execution time, and many execute but answer the wrong question. That separation is what lets a benchmark owner decide whether to improve schema retrieval, prompt constraints, value grounding, or tenant-specific adaptation.

| Downstream outcome | Count | Share of incorrect |
|--------------------|-------|--------------------|
| Answer mismatch    | 125   | 0.521              |
| Execution failed   | 79    | 0.329              |
| Schema invalid     | 25    | 0.104              |
| Parse invalid      | 11    | 0.046              |

Table 19: Downstream Text2Cypher failure taxonomy for local Qwen3.5-9B on the full exported test split. Shares exclude exact-answer matches.

| Metric       | PIPE-Cypher support                                                  | Primary citation            |
|--------------|----------------------------------------------------------------------|-----------------------------|
| ROUGE-1/2/L  | Reference overlap over serialized answer sets and query strings      | (Lin, 2004)                 |
| BLEU         | Sentence-level reference overlap with brevity penalty                | (Papineni et al., 2002)     |
| METEOR       | Exact-token METEOR-style precision/recall with fragmentation penalty | (Banerjee and Lavie, 2005)  |
| BERTScore    | Optional contextual-embedding similarity when installed              | (Zhang et al., 2020)        |
| FrugalScore  | Optional efficient learned NLG metric when installed                 | (Kamal Eddine et al., 2022) |
| Cosine       | Token-count vector cosine similarity                                 | (Manning et al., 2008)      |
| Jaro-Winkler | Character-level string similarity for near-match diagnostics         | (Jaro, 1989; Winkler, 1990) |
| Exact match  | Raw and normalized string exact match                                | (Rajpurkar et al., 2016)    |

Table 20: Supplementary reference-based text metrics supported by the PIPE-Cypher evaluator. These metrics are useful for debugging answer rendering, paraphrase sensitivity, and near-match behavior, but they do not replace execution accuracy or answer-set F1 for Text2Cypher correctness. BERTScore and FrugalScore are optional integrations because they require additional metric/model packages.

## D Diversity and Cypher Strategy Audit

### D.1 Diversity Diagnostics

PIPE-Cypher does not ask reviewers to trust a single aggregate acceptance rate. It reports diversity and strategy diagnostics so an enterprise can see whether its benchmark is balanced, what structural operators it exercises, and where residual concentration remains. Figure 10 keeps the useful diversity view: category, graph-category, and difficulty balance are strong by construction; schema and value coverage are visible audit quantities rather than hidden assumptions.

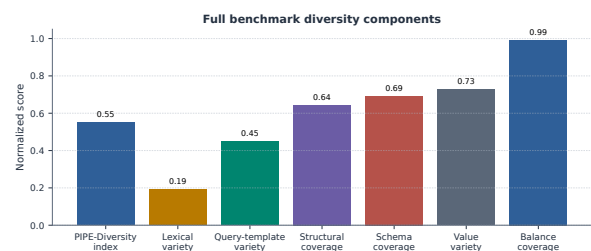


Figure 10: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the seeded-template run.

### D.2 Diversity-Governed Selection

| Metric                                  | Value                 |
|-----------------------------------------|-----------------------|
| PIPE-Diversity index                    | 0.549                 |
| Question Distinct-1                     | 0.039                 |
| Question Distinct-2                     | 0.085                 |
| Question adjusted Distinct-2            | 0.266                 |
| Question self-BLEU-2 (sampled)          | 0.813                 |
| Mean nearest-neighbor question Jaccard  | 0.775                 |
| Unique query-signature ratio            | 0.041                 |
| Top query-signature share               | 0.083                 |
| Template-family entropy                 | 0.826                 |
| Operator-combination entropy            | 0.880                 |
| Unique structural substructures         | 134                   |
| Category normalized entropy             | 1.000                 |
| Graph-category normalized entropy       | 0.980                 |
| Difficulty normalized entropy           | 0.998                 |
| Label coverage                          | 0.941                 |
| Relationship-type coverage              | 0.708                 |
| Property-name coverage                  | 0.426                 |
| Unique grounded-value ratio             | 0.357                 |
| Grounded values exactly quoted          | 0.823                 |
| Aggregation / negation / ordering rates | 0.500 / 0.125 / 0.125 |

Table 21: Diversity diagnostics for the full exported benchmark. PIPE-Diversity is a geometric mean of lexical, query-template, structural, schema, value, and balance components; component rows are shown so the composite score does not hide residual concentration.

Table 22 reports the first diversity-improvement pass: a target-50-per-graph/category selector over the 3,000 accepted full-run examples, compared with a hash-balanced random subset of the same size. The selector uses an MMR-style novelty score over query signatures, template families, structural substructures, schema atoms, entity values, and question tokens after all quality gates have already passed. It improves the PIPE-Diversity index, unique query-signature ratio, unique structural substructures, adjusted Distinct-2, and property coverage while preserving a signature-disjoint 640/80/80 split. The mixed template-family entropy and self-BLEU result is informative: it shows that post-hoc selection can improve coverage, but true diversity gains require oversampling or inducing more source templates during generation when a graph/category cell contains only one or two viable signatures.

| Metric                              | Random balanced | Diversity governed | $\Delta$ |
|-------------------------------------|-----------------|--------------------|----------|
| PIPE-Diversity index                | 0.557           | 0.575              | +0.017   |
| Unique query-signature ratio        | 0.062           | 0.135              | +0.073   |
| Top signature share (lower better)  | 0.062           | 0.062              | +0.000   |
| Template-family entropy             | 0.903           | 0.868              | -0.035   |
| Operator-combination entropy        | 0.901           | 0.911              | +0.010   |
| Unique structural substructures     | 97.000          | 134.000            | +37.000  |
| Question self-BLEU-2 (lower better) | 0.850           | 0.866              | +0.016   |
| Adjusted Distinct-2                 | 0.266           | 0.284              | +0.018   |
| Property coverage                   | 0.407           | 0.426              | +0.019   |

Table 22: Balanced subset comparison at the same graph/category target. The diversity-governed selector applies MMR-style novelty over Cypher signatures, template families, structural substructures, schema atoms, values, and question tokens after quality gates have already passed; structural/schema gains are reported alongside residual template concentration.

### D.3 Cypher Strategy Coverage

Strategy diagnostics complement category balance. Two questions can share a category while exercising different Cypher behavior; conversely, a category-balanced dataset can still miss joins, paths, negation, ranking, or bounded-result patterns. The strategy matrix and downstream strategy outcomes make the benchmark’s discriminative structure legible.

| Primary strategy | Examples | Share | Rel. patterns | Return arity | Downstream exec. acc. |
|------------------|----------|-------|---------------|--------------|-----------------------|
| Aggregation      | 1125     | 0.375 | 1.42          | 1.00         | 0.396                 |
| Join-heavy       | 375      | 0.125 | 2.33          | 2.33         | 0.000                 |
| Negation         | 375      | 0.125 | 1.89          | 2.84         | 0.000                 |
| Order/rank       | 375      | 0.125 | 1.87          | 3.41         | 0.000                 |
| Path             | 375      | 0.125 | 1.37          | 3.00         | 0.000                 |
| Single hop       | 375      | 0.125 | 1.00          | 2.33         | 0.324                 |

Table 23: Cypher strategy diagnostics over the full 3,000-example export. Strategy tags are derived from generated Cypher structure rather than from category labels; downstream execution accuracy is reported on the full held-out test split when a strategy appears there.

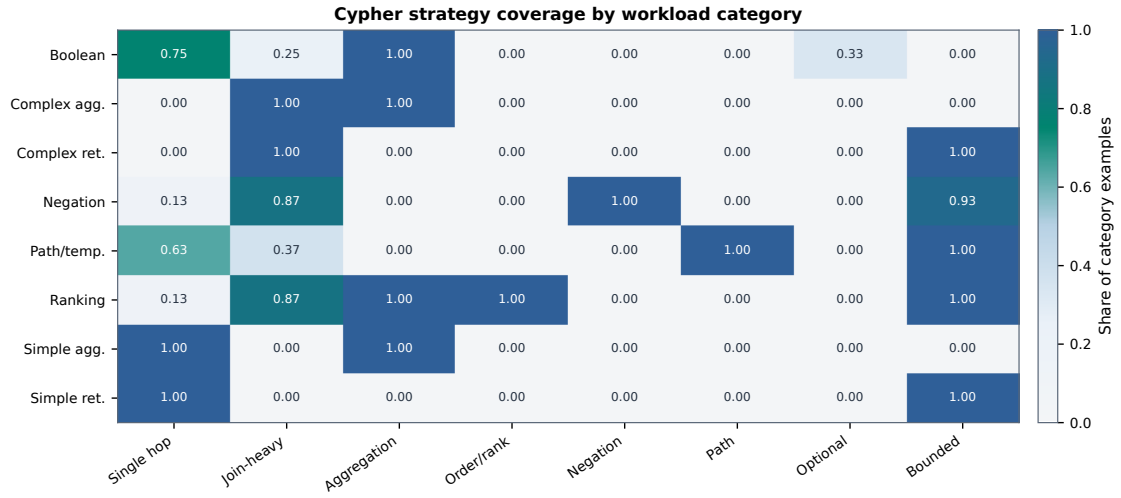


Figure 11: Cypher strategy coverage by workload category. Category balancing does not by itself guarantee operator coverage; the strategy matrix shows where the benchmark exercises single-hop retrieval, joins, aggregation, ordering, negation, path patterns, optional matches, and bounded-result queries.

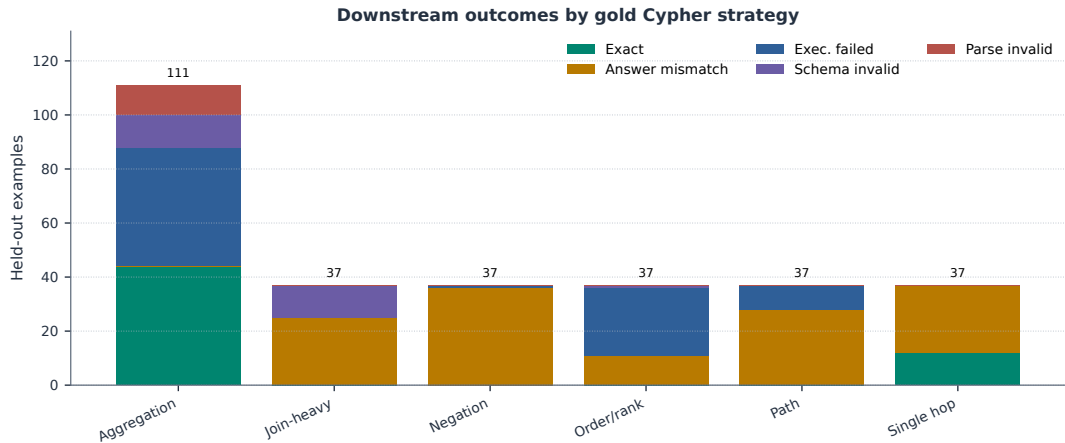


Figure 12: Downstream outcomes by gold Cypher strategy on the full held-out test split. The local Qwen3.5-9B baseline fails differently across strategies: aggregation mixes exact matches with execution failures, while join-heavy, negation, path, and ranking examples are dominated by semantically wrong executable queries or execution failures.

## E Governance, Judge Calibration, and Deployment Details

The following tables collect implementation details that support the paper’s industry claims: the validator cascade, prompt-profile controls, automation/effort contrast, and the completed post-hoc judge audit. These tables are placed here rather than near unrelated result plots because they explain how PIPE-Cypher can be deployed as a governed local workflow instead of a manual dataset-writing exercise.

### E.1 Validation Cascade

Table 24 lists the deterministic and judge gates in execution order. It is the operational view of how an enterprise deployment keeps unsafe, schema-invalid, empty, or semantically weak examples out of exported benchmarks.

| Gate or ledger             | Count | Denominator |
|----------------------------|-------|-------------|
| Export accepted            | 3,000 | 3,000       |
| Read-only safety           | 3,000 | 3,000       |
| Syntax validity            | 3,000 | 3,000       |
| Schema/value validity      | 3,000 | 3,000       |
| Execution success          | 3,000 | 3,000       |
| LLM judge pass             | 3,000 | 3,000       |
| Rejected candidates logged | 1,925 | 4,925       |

Table 24: PIPE-Cypher validation cascade for the full export and its logged candidate ledger. Unlike Mind the Query, human review is calibration-only.

### E.2 Rewrite and Governance Audits

Table 25 addresses the rewrite-preservation question directly. In the paper-generation records, generated Cypher was already identical to the normalized Cypher, so the accepted benchmark does not rely on a semantics-changing RETURN DISTINCT insertion or projection rewrite. The rewrite machinery is still part of the governed pipeline, but its paper-run role is conservative validation and logging rather than silent semantic alteration.

| Rewrite audit property              | Value |
|-------------------------------------|-------|
| Generation records audited          | 4,925 |
| Accepted records audited            | 3,000 |
| Records changed by normalization    | 0     |
| Accepted records changed            | 0     |
| RETURN DISTINCT insertions          | 0     |
| Accepted RETURN DISTINCT insertions | 0     |
| Rewrite-skip reasons logged         | 196   |
| Live comparisons required           | 0     |
| Answer-set equality in comparisons  | 0     |

Table 25: Rewrite prevalence and impact audit over paper-generation records. When no generated query differs from its normalized form, no live original/normalized re-execution is required for semantic drift.

Table 26 separates direction, schema/value, syntax/parser, and read-only issues across generation, ablation, and downstream prediction artifacts. The full generation records have no direction failures after governance; downstream predictions still contain direction errors, which is precisely the kind of graph-specific failure that a Cypher benchmark should expose.

| Evidence source         | Direction | Schema/value | Syntax/parser | Read-only |
|-------------------------|-----------|--------------|---------------|-----------|
| Full generation records | 0         | 2            | 0             | 0         |
| Target-size ablations   | 260       | 988          | 5             | 0         |
| Downstream predictions  | 25        | 9            | 12            | 0         |
| Combined                | 285       | 999          | 17            | 0         |

Table 26: Governance failure audit. Direction errors, schema/value errors, syntax/parser failures, and read-only violations are counted separately so the appendix shows which Cypher-specific gates do real work.

### E.3 Gate-Impact Counterfactual

Table 27 summarizes the first gate that blocks each non-accepted candidate. This is the counterfactual view missing from a saturated yield-only ablation: it shows what kind of bad artifact would enter the benchmark if a deployment weakened duplicate/diversity controls, non-empty execution, judge review, schema validation, direction checking, or read-only safety.



| First blocking gate | Candidates | Share |
|---------------------|------------|-------|
| Accepted            | 3,000      | 0.609 |
| Duplicate/diversity | 1,366      | 0.277 |
| Empty result        | 486        | 0.099 |
| Judge reject        | 67         | 0.014 |
| Schema              | 2          | 0.000 |
| Execution failure   | 4          | 0.001 |

Table 27: Counterfactual first-blocking-gate audit over generation records. The table shows which failure class would enter the benchmark if that gate were removed or weakened.

## E.4 Privacy and Redaction Audit

Table 28 evaluates the redaction policy rather than merely describing it. The audit builds a sensitive-value set from entity bindings, quoted Cypher literals, reverse-grounding literals, and string-valued execution samples, applies the configured redactor, and then exact-matches those raw values against the redacted question, Cypher, entity, and result surfaces. This does not replace a tenant’s PII classifier, but it gives reviewers a measurable privacy check for the value-bearing fields PIPE-Cypher itself creates.

| Redaction audit property          | Value  |
|-----------------------------------|--------|
| Examples audited                  | 3,000  |
| Sensitive values checked          | 10,956 |
| Examples with sensitive values    | 2,970  |
| Examples with residual raw values | 0      |
| Residual raw-value matches        | 0      |
| Residual rate per checked value   | 0.000  |
| Unique placeholders               | 3,754  |
| Reused placeholders               | 2,342  |
| Max placeholder frequency         | 337    |

Table 28: Exact-match redaction audit over value-bearing benchmark surfaces. The audit checks entity bindings, quoted Cypher literals, reverse grounding literals, and string-valued result samples after applying the configured redaction policy.

## E.5 Operational Accounting

Table 29 reports local-run accounting from completed generation records. These numbers are included for deployment planning: acceptance rate and graph execution latency explain throughput bottlenecks without converting the study into a paid-API cost comparison.

| Scope    | Records | Accepted | Acceptance | Exec. p50 ms | Exec. p95 ms |
|----------|---------|----------|------------|--------------|--------------|
| Overall  | 4,925   | 3,000    | 0.609      | 11.995       | 32.832       |
| FinBench | 3,405   | 2,000    | 0.587      | 11.837       | 32.398       |
| SNB      | 1,520   | 1,000    | 0.658      | 12.420       | 38.558       |

Table 29: Operational accounting from completed generation records. These are local-run latency and acceptance diagnostics, not paid-API cost claims.

## E.6 Prompt Profiles and Contracts

Table 30 documents the prompt-profile factors used for ablation planning. The full prompt contracts, including the exact LLM-judge system prompt and user prompt template used for reported judge decisions, follow immediately after the table.

| Profile                 | Added constraint                                                | Intended evidence                                     |
|-------------------------|-----------------------------------------------------------------|-------------------------------------------------------|
| Schema-only             | Use only visible schema.                                        | Baseline for schema-grounded prompting.               |
| Instructions            | Exact values, datatype rules, no nested aggregations.           | Targets MTQ-style prompt refinements.                 |
| Examples                | Placeholderized retrieved NL-Cypher pairs.                      | Tests whether examples help without extra governance. |
| Examples + instructions | Few-shot plus explicit rules.                                   | Closest controlled analogue to MTQ Table 5.           |
| Full governed           | Production-derived Cypher hints, AST-safe rewrites, judge gate. | PIPE-Cypher production setting.                       |

Table 30: Prompt profiles implemented for Mind-the-Query-style prompt-factorial evaluation. Results are reported only for completed, audited target-50-or-larger suites.

## F Prompt Contracts

PIPE-Cypher treats prompts as versioned implementation artifacts. The list summarizes the prompt contracts used for generation, repair, judging, and downstream evaluation; hashes fingerprint the full prompt constants in the codebase.

1. **Template generation.** Stage: Workload proposal. SHA-256: 10b25b.

*Contract.* Schema-only labels, relationships, properties, and categorical values; Realistic enterprise analyst wording; At most two typed slots and JSON-only output

2. **Reverse binding.** Stage: Graph grounding. SHA-256: 48e949.

*Contract.* Read-only MATCH/WHERE/RETURN DISTINCT/LIMIT only; Slot variables named exactly as requested; Forward relationship directions from the schema

3. **Cypher generation.** Stage: Candidate query. SHA-256: 61c557.

*Contract.* Only schema-visible constructs and observed directions; RETURN DISTINCT for set returns and exact equality for quoted values; Context columns, categorical hints, placeholderized retrieval, and no writes

4. **Repair.** Stage: Validation feedback. SHA-256: 56c419.

*Contract.* Preserve question intent while fixing validation or execution issues; Keep query read-only and schema-grounded; Return only corrected Cypher

5. **LLM judge.** Stage: Quality gate. SHA-256: 421c7b.

*Contract.* Inputs include question, Cypher, schema slice, execution rows, and validation summary; Strict JSON scores for ambiguity, semantic alignment, schema use, and difficulty; Categorical values constrain query literals, not observed result-row values; Pass only useful, unambiguous enterprise benchmark examples

6. **Downstream Text2Cypher.** Stage: Model evaluation. SHA-256: 4c07ff.

*Contract.* Read-only Cypher only; Schema-visible constructs and exact direction preservation; RETURN DISTINCT, count/ranking rules, and no explanations

## F.1 LLM Judge Prompt Used in Reported Runs

This is the complete LLM-judge prompt contract used for all reported judge decisions. The local judge receives a JSON-only system instruction and the following user prompt template after schema slicing, execution sampling, and deterministic validation. The placeholders are filled with the candidate question, Cypher, schema slice, execution rows, and validation summary for each reviewed example.

System prompt:  
You are an expert benchmark engineer. Return strict JSON only. No markdown or extra text.

User prompt template:  
You are judging whether an NL-to-Cypher benchmark example is acceptable for an enterprise benchmark.

Graph schema:  
{schema}

Question:  
{question}

Cypher:  
{cypher}

Execution sample:  
{rows}

Validation summary:  
{validation}

Return strict JSON with:  
- pass: boolean  
- ambiguity\_score: number from 0 to 1, lower is better  
- semantic\_alignment\_score: number from 0 to 1  
- schema\_use\_score: number from 0 to 1

- difficulty: one of easy, medium, hard  
- failure\_reason: short string, empty if pass is true

Pass only if the question is unambiguous, the Cypher answers it, the schema use is valid, and the result would be useful in an enterprise benchmark.  
- Categorical property values in the schema constrain literal values written in the Cypher query.  
- Do not reject because the execution sample returns a value that is absent from the categorical-value list; result rows are observed graph outputs.  
- If deterministic validation says schema use is valid, lower schema\_use\_score only when the Cypher itself uses nonexistent schema elements, invalid relationship directions, or invalid literal values.

## F.2 Automation and Calibration

Table 31 explains the industry deployment contrast with Mind the Query, and Table 32 reports the completed human calibration packet. The important claim is not that human review disappeared from science; it is that human labels are post-hoc calibration evidence rather than a required production gate.

| Dimension              | Mind the Query                    | PIPE-Cypher                               |
|------------------------|-----------------------------------|-------------------------------------------|
| Generation review gate | Manual logical review             | Deterministic gates + local LLM judge     |
| Human effort           | Reported 1,400 person-hours       | 80-row post-hoc judge calibration audit   |
| Private values         | Public benchmark values           | Configurable sampling and redacted export |
| Refresh                | Static dataset release            | Rerunnable private benchmark factory      |
| Model endpoint         | Gemini in their reported pipeline | Local Qwen3.5-9B endpoint                 |

Table 31: Industry deployment contrast with Mind the Query. PIPE-Cypher focuses on private refreshable benchmark generation rather than a one-time public dataset.

| Audit packet property      | Value               |
|----------------------------|---------------------|
| Rows                       | 80                  |
| Judge accept / reject      | 40 / 40             |
| FinBench / SNB rows        | 48 / 32             |
| Easy / medium rows         | 26 / 54             |
| Labeled rows               | 80                  |
| Calibration status         | complete            |
| Agreement / $\kappa$       | 0.800 / 0.600       |
| Judge precision (95% CI)   | 1.000 (0.912–1.000) |
| Judge recall (95% CI)      | 0.714 (0.585–0.816) |
| False-accept rate (95% CI) | 0.000 (0.000–0.138) |
| False-reject rate (95% CI) | 0.286 (0.184–0.415) |

Table 32: Post-hoc judge calibration packet coverage and agreement. Human labels calibrate the automated gate after generation and are not used as a generation gate.

| Review concern              | Response in revision                                                                                                              | Evidence artifact                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Missing tables/code package | Add package verifier and anonymous bundle builder; require all LaTeX inputs, code roots, and evidence manifests.                  | Submission package manifest and verifier report. |
| Novelty under-specified     | Recast novelty as benchmark-factory governance rather than execution validation alone.                                            | Table 1.                                         |
| Judge calibration limited   | Keep 80-row audit as the only completed evidence; larger packet is an artifact until labeled.                                     | Table 32.                                        |
| Few-shot leakage            | Make no-signature control the generalization result; same-signature banks are operational upper bounds.                           | Tables 16–18.                                    |
| Overbroad enterprise claim  | State that enterprise onboarding is designed and evaluated on three public proxy graphs, not proven on arbitrary private tenants. | Table 13 and ICIJ audit.                         |
| Saturated ablations         | Interpret ablations as reliability of the grounded governance core; use gates/failure taxonomy for component evidence.            | Tables 10, 27.                                   |
| Privacy not evaluated       | Add exact-match redaction audit over value-bearing surfaces and canary tests.                                                     | Table 28.                                        |

Table 33: Review-response matrix. Each major review concern is addressed by either new evidence, narrowed claims, or an explicit limitation.

## G Claim–Evidence Map

This section maps the main paper claims to the strongest current evidence and to the remaining risks. This is intended as a reviewer-facing audit surface: claims with pending evidence remain marked as such rather than being folded into the main results.

1. **Claim 1.** PIPE-Cypher can generate a private, executable NL-to-Cypher benchmark over enterprise-style property graphs using local models.

*Evidence.* The full local Qwen3.5-9B run exported 3,000 accepted examples: 2,000 FinBench and 1,000 SNB, with 375 examples in every planned workload category and all accepted examples passing read-only, syntax, schema, execution, non-empty-result, and judge gates.

*Key artifacts.* benchmark export; benchmark export; snapshot: clean qwen9b submission evidence manifest; paper table: benchmark export; +1 more

*Status.* Supported by live local-model evidence.

*Risk.* Generation quality may change with private enterprise schemas beyond FinBench/SNB.

2. **Claim 2.** Cypher-specific governance makes generated benchmark candidates safer and more auditable than prompt-only generation.

*Evidence.* The pipeline validates read-only safety, schema-visible labels, node and relationship properties, relationships, observed relationship direction, categorical values, contextual return columns, parser-style structure, execution success, and LLM-judge review. In the clean full run, only 2 of 4,925 candidates were schema-invalid after deterministic governance, and all 3,000 exported examples passed the deterministic and judge gates.

*Key artifacts.* code: validator.py; code: cypher\_parser.py; code: question\_constraints.py; snapshot: failure taxonomy; +1 more

*Status.* Supported by code, tests, and full-run failure taxonomy.

*Risk.* Semantic correctness still depends on execution evidence and judge review; the completed 80-row audit is a calibration sample rather than an exhaustive proof.

3. **Claim 3.** Reverse grounding and structured workload seeds are designed to improve reliability under local-model constraints.

*Evidence.* Three FinBench/SNB ablation suites are complete and evidence-ready: the original target-50 suite, the corrected target-100 suite, and a seed-17 target-50 repeat. Each suite completed 14/14 graph/variant cells, reached every category target for all non-empty/non-unconstrained runs, and passed paper-readiness audits with logs, model IDs, code revisions, run summaries, collection manifests, and gate-rate availability. The target-100 suite is the detailed appendix ablation table/figure, while the three-suite comparison reports target-normalized coverage, acceptance variation, execution rates, and judge rates for stability.

*Key artifacts.* script: run live ablation suite; script: monitor remote ablation suite; script: monitor remote ablation queue; script: collect remote ablation suite; +20 more

*Status.* Supported by audited target-100 evidence and three-suite target-size/repeated-seed comparison.

*Risk.* The repaired unconstrained local generation baseline produced accepted examples only in a narrow subset of categories: 200/422 accepted on FinBench across 2/8 categories and 50/2,000 accepted on SNB across 0/8 category targets. Comparisons against it should be framed as a gate-stress baseline rather than a tuned generation system.

4. **Claim 4.** The benchmark controls category and difficulty balance while exposing residual diversity limits.

*Evidence.* The full export has perfect category balance, planned 2:1 FinBench/SNB graph mix, and a near-even easy/medium split. Diversity diagnostics report Distinct-n, sampled self-BLEU-2, query-signature diversity, normalized entropy, schema coverage, structural feature rates, and aggregate value-grounding diagnostics. The full export uses 1,115 unique grounded entity values and exactly quotes

|      |                                                         |      |  |
|------|---------------------------------------------------------|------|--|
| 1069 | grounded values in 82.6% of examples with               |      |  |
| 1070 | entity bindings, making template and value              |      |  |
| 1071 | concentration visible instead of hidden.                |      |  |
| 1072 | <i>Key artifacts.</i> snapshot: diversity report; snap- |      |  |
| 1073 | shot: diversity improvement comparison; pa-             |      |  |
| 1074 | per table: diversity; paper table: diversity im-        |      |  |
| 1075 | provement; +2 more                                      |      |  |
| 1076 | <i>Status.</i> Supported by full-export statistics and  |      |  |
| 1077 | diversity diagnostics.                                  |      |  |
| 1078 | <i>Risk.</i> Low query-signature diversity in the       |      |  |
| 1079 | reported local-model run remains a limitation           |      |  |
| 1080 | and ablation target.                                    |      |  |
| 1081 | 5. <b>Claim 5.</b> The generated benchmark is useful    |      |  |
| 1082 | for downstream Text2Cypher evaluation.                  |      |  |
| 1083 | <i>Evidence.</i> An 11-model completed local            |      |  |
| 1084 | downstream evaluation over the 296-example              |      |  |
| 1085 | full test split reports parse validity, schema          |      |  |
| 1086 | validity, execution success, execution accu-            |      |  |
| 1087 | racy, answer F1, per-category breakdowns,               |      |  |
| 1088 | model-transfer summaries, few-shot control              |      |  |
| 1089 | runs, leakage audits, and row-level error               |      |  |
| 1090 | taxonomies. Zero-shot execution accuracy                |      |  |
| 1091 | ranges from 0.000 to 0.203 with a mean of               |      |  |
| 1092 | 0.036. Few-shot controls over the same split            |      |  |
| 1093 | improve the mean to 0.200 for a scored no-              |      |  |
| 1094 | signature condition that excludes exact query-          |      |  |
| 1095 | signature matches and near-duplicate ques-              |      |  |
| 1096 | tions, and to 0.269/0.267 for ordered/random            |      |  |
| 1097 | same-category example banks. Model-level                |      |  |
| 1098 | paired bootstrap intervals show that gains are          |      |  |
| 1099 | concentrated in compatible model families               |      |  |
| 1100 | rather than universal.                                  |      |  |
| 1101 | <i>Key artifacts.</i> downstream summary; snap-         |      |  |
| 1102 | shot: downstream control manifest; snapshot:            |      |  |
| 1103 | model transfer summary; snapshot: fewshot               |      |  |
| 1104 | control summary; +12 more                               |      |  |
| 1105 | <i>Status.</i> Supported by live downstream evalua-     |      |  |
| 1106 | tion against FinBench/SNB databases, with               |      |  |
| 1107 | complete 11-model zero-shot/control sum-                |      |  |
| 1108 | maries and leakage-aware few-shot controls.             |      |  |
| 1109 | <i>Risk.</i> Several local fine-tuned checkpoints re-   |      |  |
| 1110 | main brittle under few-shot prompting; the              |      |  |
| 1111 | current study shows example-bank utility and            |      |  |
| 1112 | transfer failure modes, not a completed tenant-         |      |  |
| 1113 | specific fine-tuning result. The stricter no-           |      |  |
| 1114 | signature improvement is an aggregate signal            |      |  |
| 1115 | with broad model-family uncertainty rather              |      |  |
| 1116 | than a universal model-level gain.                      |      |  |
|      | 6. <b>Claim 6.</b> Automated LLM-judge review can       | 1117 |  |
|      | serve as a generation gate with post-hoc hu-            | 1118 |  |
|      | man calibration while still exposing judge              | 1119 |  |
|      | risk.                                                   | 1120 |  |
|      | <i>Evidence.</i> The pipeline uses deterministic val-   | 1121 |  |
|      | idation plus LLM-judge review as the gener-             | 1122 |  |
|      | ation gate. A post-hoc 80-row judge audit               | 1123 |  |
|      | packet preserves 40 judge accepts, 40 judge             | 1124 |  |
|      | rejects, both graphs, and all eight categories.         | 1125 |  |
|      | The completed human audit reports 80.0%                 | 1126 |  |
|      | agreement, Cohen’s kappa 0.60, balanced ac-             | 1127 |  |
|      | curacy 0.857, judge precision 1.00, judge               | 1128 |  |
|      | specificity 1.00, judge recall 0.714, zero false        | 1129 |  |
|      | accepts, and 16 false rejects. The analyzer             | 1130 |  |
|      | requires complete labels for paper-facing cali-         | 1131 |  |
|      | bration, and the tracked summary avoids com-            | 1132 |  |
|      | mitting raw value-bearing audit rows.                   | 1133 |  |
|      | <i>Key artifacts.</i> code: judge.py; code: calibra-    | 1134 |  |
|      | tion.py; code: judge_audit_packet.py; script:           | 1135 |  |
|      | create judge annotation sheets; +3 more                 | 1136 |  |
|      | <i>Status.</i> Supported by completed human audit       | 1137 |  |
|      | summary.                                                | 1138 |  |
|      | <i>Risk.</i> The judge appears conservative on this     | 1139 |  |
|      | sample; larger audits or different enterprise           | 1140 |  |
|      | schemas may reveal false accepts.                       | 1141 |  |
|      | 7. <b>Claim 7.</b> PIPE-Cypher supports arbitrary en-   | 1142 |  |
|      | terprise schema onboarding with configurable            | 1143 |  |
|      | privacy redaction and value-sampling poli-              | 1144 |  |
|      | cies.                                                   | 1145 |  |
|      | <i>Evidence.</i> The repo includes an enterprise        | 1146 |  |
|      | deployment guide, an enterprise template con-           | 1147 |  |
|      | fig, privacy config fields, schema introspec-           | 1148 |  |
|      | tion that reads categorical-value thresholds,           | 1149 |  |
|      | maximum sampled value length, and omitted-              | 1150 |  |
|      | property patterns from config, deterministic            | 1151 |  |
|      | redaction helpers, and a redacted benchmark             | 1152 |  |
|      | export CLI. The redaction policy replaces               | 1153 |  |
|      | quoted Cypher literals, question mentions,              | 1154 |  |
|      | entity values, and string-valued result sam-            | 1155 |  |
|      | ples with stable placeholders while preserving          | 1156 |  |
|      | query structure and gate metadata. The repo             | 1157 |  |
|      | now also includes an ICIJ Offshore Leaks                | 1158 |  |
|      | onboarding profile, config, download/load               | 1159 |  |
|      | helpers, graph plan, schema-derived sparse-             | 1160 |  |
|      | category templates, and a completed sanitized           | 1161 |  |
|      | target-100 onboarding audit on a public fi-             | 1162 |  |
|      | nance/compliance graph beyond the LDBC                  | 1163 |  |
|      | study workloads.                                        | 1164 |  |
|      | <i>Key artifacts.</i> research note: enterprise de-     | 1165 |  |

ployment guide; research note: icij offshore-leaks graph plan; enterprise\_template.yaml; schema\_icij\_offshoreleaks.json; +21 more

*Status.* Supported by docs, config, code, deterministic tests, a concrete third-graph onboarding plan, and a completed sanitized ICIJ target-100 live onboarding run with 800/983 accepted examples and 8/8 categories at target.

*Risk.* ICIJ is public rather than private/anonymized enterprise data. The strongest industry validation would still come from a real tenant graph under the same audit protocol.

## H Representative Accepted Examples

The examples below are selected in stable identifier order from the tracked full-export snapshot, one per graph/category cell when available. They show the natural-language question, accepted Cypher, structural tags, gate status, and a bounded execution-result sample.

1. **FinBench / Boolean Existence / easy. Question:** Does account '187743809466009406' have any outgoing transfer?

```
MATCH (src:Account {accountId:
'187743809466009406'})
-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT COUNT(DISTINCT src)
> 0
AS HasOutgoingTransfer
```

*Structure:* single\_hop, aggregation.

*Relationships:* TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {HasOutgoingTransfer: True}; observed rows: 1

2. **FinBench / Complex Aggregation / medium. Question:** What is the total transferred amount from accounts owned by person 'Gwar'?

```
MATCH (p:Person {personName:
'Gwar'})
-[:OWN_ACCOUNT]->
(src:Account)-[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT SUM(t.amount) AS
TotalTransferredAmount
```

*Structure:* join\_heavy, aggregation.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {TotalTransferredAmount: 14972318.41}; observed rows: 1

3. **FinBench / Complex Retrieval / easy. Question:** Which accounts received transfers from accounts owned by person 'Zof'?

```
MATCH (p:Person {personName:
'Zof'})
-[:OWN_ACCOUNT]-> (src:Account)
-[:TRANSFER_TO]-> (dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4687402787162554854, AccountType: credit card, IsBlocked: False}; observed rows: 3

4. **FinBench / Negation Difference / medium. Question:** Which accounts owned by person 'Kant' have not sent any transfers?

```
MATCH (p:Person {personName:
'Kant'})
-[:OWN_ACCOUNT]-> (a:Account)
WHERE NOT (a) -[:TRANSFER_TO]->
(:Account)
RETURN DISTINCT a.accountId AS
AccountId, a.accountType AS
AccountType,
a.isBlocked AS IsBlocked
LIMIT 300
```

*Structure:* join\_heavy, negation, bounded\_result.

*Relationships:* OWN\_ACCOUNT, TRANSFER\_TO.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId: 4739757132830738341, AccountType: merchant account, IsBlocked: False}; observed rows: 1

5. **FinBench / Path Temporal / medium. Question:** Which accounts can receive money within two transfer hops from accounts owned by person 'Sossamon'?

```
MATCH (p:Person {personName:
'Sossamon'}) -[:OWN_ACCOUNT]->
(src:Account) -[:TRANSFER_TO*1..2]->
(dst:Account)
RETURN DISTINCT dst.accountId AS
AccountId, dst.accountType AS
AccountType, dst.isBlocked AS
IsBlocked
LIMIT 300
```



1273 *Structure:* join\_heavy, path, bounded\_result.

1274 *Relationships:* OWN\_ACCOUNT, TRANS-

1275 *FER\_TO.*

1276 *Gates:* RO/Syn/Schema/Exec/Judge.

1277 *Result sample:* {AccountId:

1278 4687402787162554854, AccountType:

1279 credit card, IsBlocked: False}; observed rows:

1280 4

- 1281 **6. FinBench / Ranking Topk / medium. Ques-**
- 1282 *tion:* For accounts owned by person 'Barry',
- 1283 which account sent the highest total transfer
- 1284 amount?

1285 

```
MATCH (p:Person {personName:
1286 'Barry'})
1287 -[:OWN_ACCOUNT]->
1288 (src:Account)-[t:TRANSFER_TO]->
1289 (:Account)
1290 WITH src, SUM(t.amount) AS
1291 totalAmount
1292 RETURN DISTINCT src.accountId AS
1293 AccountId, src.accountType AS
1294 AccountType, src.isBlocked AS
1295 IsBlocked,
1296 totalAmount
1297 ORDER BY totalAmount DESC
1298 LIMIT 1
```

1299 *Structure:* join\_heavy, aggregation, or-

1300 der\_rank, bounded\_result.

1301 *Relationships:* OWN\_ACCOUNT, TRANS-

1302 *FER\_TO.*

1303 *Gates:* RO/Syn/Schema/Exec/Judge.

1304 *Result sample:* {AccountId:

1305 4732438783436260772, AccountType:

1306 internet account, IsBlocked: False}; observed

1307 rows: 1

- 1308 **7. FinBench / Simple Aggregation / easy. Ques-**
- 1309 *tion:* How many accounts are owned by per-
- 1310 son 'Kaewsuktae'?

1311 

```
MATCH (p:Person {personName:
1312 'Kaewsuktae'}) -[:OWN_ACCOUNT]->
1313 (a:Account)
1314 RETURN DISTINCT COUNT(DISTINCT a) AS
1315 AccountCount
```

1316 *Structure:* single\_hop, aggregation.

1317 *Relationships:* OWN\_ACCOUNT.

1318 *Gates:* RO/Syn/Schema/Exec/Judge.

1319 *Result sample:* {AccountCount: 1}; observed

1320 rows: 1

- 1321 **8. FinBench / Simple Retrieval / easy. Ques-**
- 1322 *tion:* Which accounts are owned by person
- 1323 'Barry'?

```
MATCH (p:Person {personName:
1324 'Barry'})
1325 -[:OWN_ACCOUNT]-> (a:Account)
1326 RETURN DISTINCT a.accountId AS
1327 AccountId, a.accountType AS
1328 AccountType,
1329 a.isBlocked AS IsBlocked
1330 LIMIT 300
1331
```

*Structure:* single\_hop, bounded\_result.

*Relationships:* OWN\_ACCOUNT.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {AccountId:

4732438783436260772, AccountType:

internet account, IsBlocked: False}; observed

rows: 1

- 9. SNB / Boolean Existence / medium. Ques-**
- tion:* Does person with id 6597069766828
- like any post?

```
MATCH (p:Person {id:
1342 6597069766828})
1343 OPTIONAL MATCH (p) -[:LIKES]->
1344 (post:Post)
1345 RETURN DISTINCT COUNT(post) > 0 AS
1346 LikesAnyPost
1347
```

*Structure:* single\_hop, aggregation, optional.

*Relationships:* LIKES.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {LikesAnyPost: True}; ob-

erved rows: 1

- 10. SNB / Complex Aggregation / medium.**
- Question:* How many distinct posts
- are in forums joined by person with id
- 6597069766845?

```
MATCH (forum:Forum) -[:HAS_MEMBER]->
1357 (p:Person {id: 6597069766845})
1358 MATCH (forum) -[:CONTAINER_OF]->
1359 (post:Post)
1360 RETURN DISTINCT COUNT(DISTINCT post)
1361 AS
1362 JoinedForumPostCount
1363
```

*Structure:* join\_heavy, aggregation.

*Relationships:* CONTAINER\_OF,

HAS\_MEMBER.

*Gates:* RO/Syn/Schema/Exec/Judge.

*Result sample:* {JoinedForumPostCount: 1};

observed rows: 1

- 11. SNB / Complex Retrieval / easy. Question:**
- Which people are members of forums contain-
- ing posts tagged 'Manuel\_Noriega'?

```
MATCH (forum:Forum) -[:HAS_MEMBER]->
1373 (p:Person), (forum)
1374 -[:CONTAINER_OF]->
1375 (post:Post) -[:HAS_TAG]-> (tag:Tag
1376 {name: 'Manuel_Noriega'})
1377 RETURN DISTINCT p.id AS PersonId
1378 LIMIT 200
1379
```

|      |                                                        |                                                       |      |
|------|--------------------------------------------------------|-------------------------------------------------------|------|
| 1380 | <i>Structure:</i> join_heavy, bounded_result.          | <i>Relationships:</i> IS_LOCATED_IN.                  | 1433 |
| 1381 | <i>Relationships:</i> CONTAINER_OF,                    | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.               | 1434 |
| 1382 | HAS_MEMBER, HAS_TAG.                                   | <i>Result sample:</i> {TargetId: 164, TargetName:     | 1435 |
| 1383 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                | Kolkata, relatedCount: 130}; observed rows:           | 1436 |
| 1384 | <i>Result sample:</i> {PersonId: 4398046511220};       | 10                                                    | 1437 |
| 1385 | observed rows: 19                                      |                                                       |      |
| 1386 | <b>12. SNB / Negation Difference / easy. Question:</b> | <b>15. SNB / Simple Aggregation / easy. Question:</b> | 1438 |
| 1387 | Which person records are not linked from any           | How many posts are tagged 'Vietnam'?                  | 1439 |
| 1388 | message record through :HAS_CREATOR?                   |                                                       |      |
| 1389 | <pre>MATCH (p:Person)</pre>                            | <pre>MATCH (post:Post) -[:HAS_TAG]-&gt;</pre>         | 1440 |
| 1390 | <pre>WHERE NOT EXISTS((:Message)</pre>                 | <pre>(tag:Tag</pre>                                   | 1441 |
| 1391 | <pre>-[:HAS_CREATOR]-&gt; (p))</pre>                   | <pre>{name: 'Vietnam'})</pre>                         | 1442 |
| 1392 | <pre>RETURN DISTINCT p.id AS PersonId,</pre>           | <pre>RETURN DISTINCT COUNT(DISTINCT post)</pre>       | 1443 |
| 1393 | <pre>p.firstName AS PersonFirstName,</pre>             | <pre>AS</pre>                                         | 1444 |
| 1394 | <pre>p.lastName AS PersonLastName</pre>                | <pre>PostCount</pre>                                  | 1445 |
| 1395 | <i>Structure:</i> single_hop, negation.                | <i>Structure:</i> single_hop, aggregation.            | 1446 |
| 1396 | <i>Relationships:</i> HAS_CREATOR.                     | <i>Relationships:</i> HAS_TAG.                        | 1447 |
| 1397 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.               | 1448 |
| 1398 | <i>Result sample:</i> {PersonFirstName: R., Per-       | <i>Result sample:</i> {PostCount: 1}; observed        | 1449 |
| 1399 | sonId: 8796093022313, PersonLastName:                  | rows: 1                                               | 1450 |
| 1400 | Rao}; observed rows: 25                                |                                                       |      |
| 1401 | <b>13. SNB / Path Temporal / easy. Question:</b>       | <b>16. SNB / Simple Retrieval / easy. Question:</b>   | 1451 |
| 1402 | Which people are within two knows hops of              | Which post IDs did person with id                     | 1452 |
| 1403 | person with id 4398046511136?                          | 4398046511124 like?                                   | 1453 |
| 1404 | <pre>MATCH (src:Person {id:</pre>                      | <pre>MATCH (p:Person {id:</pre>                       | 1454 |
| 1405 | <pre>4398046511136})</pre>                             | <pre>4398046511124})</pre>                            | 1455 |
| 1406 | <pre>-[:KNOWS*1..2]-&gt; (dst:Person)</pre>            | <pre>-[:LIKES]-&gt; (post:Post)</pre>                 | 1456 |
| 1407 | <pre>RETURN DISTINCT dst.id AS PersonId,</pre>         | <pre>RETURN DISTINCT post.id AS PostId</pre>          | 1457 |
| 1408 | <pre>dst.firstName AS FirstName,</pre>                 | <pre>LIMIT 200</pre>                                  | 1458 |
| 1409 | <pre>dst.lastName</pre>                                |                                                       |      |
| 1410 | <pre>AS LastName</pre>                                 | <i>Structure:</i> single_hop, bounded_result.         | 1459 |
| 1411 | <pre>LIMIT 200</pre>                                   | <i>Relationships:</i> LIKES.                          | 1460 |
| 1412 | <i>Structure:</i> single_hop, path, bounded_result.    | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.               | 1461 |
| 1413 | <i>Relationships:</i> KNOWS.                           | <i>Result sample:</i> {PostId: 343597385744}; ob-     | 1462 |
| 1414 | <i>Gates:</i> RO/Syn/Schema/Exec/Judge.                | served rows: 5                                        | 1463 |
| 1415 | <i>Result sample:</i> {FirstName: Rafael,              |                                                       |      |
| 1416 | LastName: Fernández, PersonId:                         |                                                       |      |
| 1417 | 4398046511333}; observed rows: 25                      |                                                       |      |
| 1418 | <b>14. SNB / Ranking Topk / medium. Question:</b>      |                                                       |      |
| 1419 | Which city records are linked                          |                                                       |      |
| 1420 | from the most organisation records through             |                                                       |      |
| 1421 | :IS_LOCATED_IN?                                        |                                                       |      |
| 1422 | <pre>MATCH (s:Organisation)</pre>                      |                                                       |      |
| 1423 | <pre>-[:IS_LOCATED_IN]-&gt; (e:City)</pre>             |                                                       |      |
| 1424 | <pre>WITH e, COUNT(DISTINCT s) AS</pre>                |                                                       |      |
| 1425 | <pre>relatedCount</pre>                                |                                                       |      |
| 1426 | <pre>RETURN DISTINCT e.id AS TargetId,</pre>           |                                                       |      |
| 1427 | <pre>e.name</pre>                                      |                                                       |      |
| 1428 | <pre>AS TargetName, relatedCount</pre>                 |                                                       |      |
| 1429 | <pre>ORDER BY relatedCount DESC</pre>                  |                                                       |      |
| 1430 | <pre>LIMIT 10</pre>                                    |                                                       |      |
| 1431 | <i>Structure:</i> single_hop, aggregation, or-         |                                                       |      |
| 1432 | der_rank, bounded_result.                              |                                                       |      |